TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Knitted Samples to a Given Specification and Their Dimensional Behavior

Knit to a target — areal mass and dimensions in the fully relaxed state — then follow the samples through relaxation and washing: shrinkage, growth, spirality, and the dimensional constants that let the next specification be engineered rather than guessed.

FM-4202 · ADVANCED KNITTING SESSIONAL · EXPERIMENT 5 · FABRIC MANUFACTURING SPECIALIZATION

1. Aim and theory

Aim: to produce single-jersey samples meeting a stated specification (yarn, target GSM and dimensions in the fully relaxed state), and to quantify their dimensional change through dry relaxation, wet relaxation and washing, including spirality.

A knitted fabric's dimensions are not fixed at the machine: the loops leave the needles strained into the machine's geometry and then **relax** toward their low-energy shape, contracting (usually) in both directions, in stages — **dry relaxation** (time off the machine), **wet relaxation** (soaking lets fibres slip to the new shape), and the **fully relaxed state** after repeated wash–dry cycles, the only state in which a specification is honest. Two anchors make the behaviour predictable (TE-3205): the **stitch length l does not change** through all of it — only loop shape does — and in the fully relaxed state the products CPC × l and WPC × l settle to near-constant values for a given yarn and structure (the dimensional constants of Munden's analysis). Spirality — the skewing of courses against wales in single jersey — rides along with relaxation as the yarn's residual twist torque reasserts itself in the loop.

2. Apparatus and materials

- Single-jersey machine of Experiments 1–3; specified yarn (19.7 tex cotton in the model run); template and indelible marker for gauge marks (50 cm × 50 cm reference square with three gauge lines each way); measuring table and steel rule; wash wheel or domestic machine with standard programme, mild detergent; flat drying racks; GSM cutter and balance; tumble dryer for the wash–dry cycles.

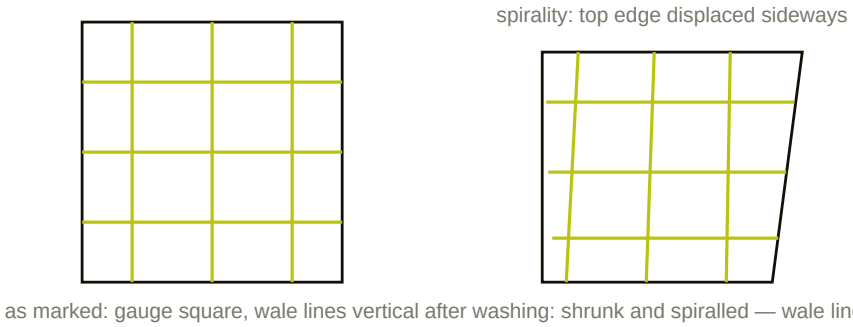


Figure 1 — The gauge-mark method: indelible lines in both directions are measured at every state; in the washed specimen the wale-direction lines lean (spirality) as well as closing up (shrinkage). Dimensions and skew are read from the same marks.

3. Procedure

1. **Engineer the sample:** from the specification (fully relaxed GSM 150 ± 5 g/m², 19.7 tex cotton jersey), choose the stitch length via the tightness factor: K = √tex ÷ l; for K ≈ 1.39, l = 4.44 ÷ 1.39 ≈ 3.2 mm. Set the machine to deliver l = 3.2 mm (course-length verification, Experiment 1) and knit the sample length.
2. **Mark:** on the dry-relaxed sample, mark the 50 × 50 cm gauge square with three lines each way, clear of edges.
3. **States in sequence,** measuring all gauge lines (and GSM) at each: (a) on-machine/grey as knitted; (b) dry-relaxed 24 h flat; (c) wet-relaxed (soak 30 min, flat dry); (d) after 1 wash–dry cycle; (e) after 3 and 5 cycles (the approach to fully relaxed).
4. **Spirality:** at the fully relaxed state, measure the sideways displacement of a wale gauge line over its length; compute spirality % and angle.
5. **Constants:** at the final state compute CPC × l, WPC × l and K; test the GSM identity one last time; compare with Experiment 3's dry-state values and explain the differences.

4. Worked calculation (model data)

SHRINKAGE AND SPIRALITY OF THE MODEL SAMPLE, GREY → FULLY RELAXED

Gauge length (course direction): 50.0 cm grey → 46.5 cm fully relaxed: shrinkage = (50.0 − 46.5) ÷ 50.0 × 100 = **7.0% lengthwise**.

Gauge width (wale direction): 50.0 → 47.8 cm: shrinkage = (50.0 − 47.8) ÷ 50.0 × 100 = **4.4% widthwise**.

Spirality: wale line displaced 6.0 cm sideways over 50 cm: spirality = 6.0 ÷ 50 × 100 = **12%**; angle = $\tan^{-1}(6 \div 50)$ = **6.8°**.

Constants at the final state (CPC 17/cm, WPC 11.5/cm, l = 3.2 mm — l unchanged, as theory

requires): CPC × l = 17 × 3.2 = **54.4**; WPC × l = 11.5 × 3.2 = **36.8**; and the identity: GSM = 11.5 × 17 × 3.2 × 19.7 ÷ 100 ≈ **123 g/m²** — this sample finished *below* its 150 g/m² target, the instructive failure analysed in Section 6.

5. Observation table (model data)

STATE	LENGTH (CM)	WIDTH (CM)	CPC /CM	WPC /CM	GSM (G/M²)
Grey (off machine)	50.0	50.0	15.8	11.0	110
Dry-relaxed 24 h	48.6	49.1	16.3	11.2	115
Wet-relaxed	47.4	48.3	16.7	11.4	120
After 1 wash	46.8	47.9	16.9	11.5	122
After 5 washes (fully relaxed)	46.5	47.8	17.0	11.5	123

Every row satisfies the identity GSM = WPC × CPC × l × tex ÷ 100 at l = 3.2 mm, 19.7 tex — the densities are the gauge measurements restated, and the mass follows. (CPC and WPC at each state scale inversely with the gauge length and width, as they must.)

Most movement happens early (dry + wet relaxation deliver the bulk); the last cycles converge — the operational definition of "fully relaxed" is that further cycles stop changing the numbers.

6. Discussion — the missed target

The sample was engineered for 150 g/m² and delivered 123. The post-mortem is the experiment's real content: the design used a tightness factor (K = 1.39, l = 3.2 mm) but took its expected CPC/WPC from a *dry-relaxed* reference; washing carried CPC to 17 and WPC to 11.5, whose product with l and tex gives 123. To hit 150 in the fully relaxed state the design must run *backwards from fully relaxed constants*: with l = 3.2 and 19.7 tex, GSM 150 needs WPC × CPC = 150 × 100 ÷ (3.2 × 19.7) ≈ 238 — e.g. CPC 17 × WPC 14.0, a WPC this yarn/gauge combination will not relax to at l = 3.2; the honest corrections are a shorter stitch length (l ≈ 2.9 mm raises both densities' product and K together) — recompute, re-knit, re-measure. Specifications are met in the state they are written for, or they are not met at all.

7. Precautions

- Gauge marks with a template, indelible after washing; frayed or distorted edge zones are never measured.
- Dry flat, without tension, at every state; a sample dried hanging has been measured in a state nobody specified.
- Same wash programme, load and detergent throughout; a changed cycle is a changed experiment.
- Report the state beside every number — a CPC without its state is not data (the lesson of the missed target).

8. Practice (viva preparation)

1. A specimen's gauge width goes 50.0 → 48.0 cm and length 50.0 → 45.5 cm. Compute both shrinkages and say which direction relaxed more.
2. Why does stitch length stay constant through relaxation while CPC changes?
3. A fully relaxed jersey has CPC 18/cm at l = 3.0 mm. Compute CPC × l and state what the constant is used for.
4. Give two independent controls against spirality (one at yarn stage, one at finishing).
5. State why specifications for knitted fabrics are written in the fully relaxed state.

Answer key

1. Width: (50.0 − 48.0)/50.0 = **4.0%**; length: (50.0 − 45.5)/50.0 = **9.0%** — lengthwise (course direction) relaxed more, the usual single-jersey pattern.
2. Stitch length is the amount of yarn in the loop — a material quantity that relaxation cannot create or destroy. Relaxation changes the loop's *shape* (rounder, shorter in the course direction), so more courses pack per centimetre: CPC rises at constant l. That is precisely why l is the machine setting that survives to the finished fabric.
3. CPC × l = 18 × 3.0 = **54** — the dimensional constant: for this yarn/structure in the fully relaxed state it is reproducible, so a designer who fixes l can predict the relaxed CPC (and, with its WPC partner, the GSM) before knitting.
4. Yarn stage: **twist balance** — use plied yarns or alternate S/Z twist feeders, or steam-set the yarn to kill residual torque. Finishing: **heat-setting** (synthetics) or resin/compacting that locks the relaxed square geometry (cotton).
5. Because it is the only state that is (a) reproducible between laboratories and (b) the state the garment actually lives in after its first washes. A spec written in the grey state certifies a fabric the customer will never meet.
